# Battery Management System User Manual – Zips Racing

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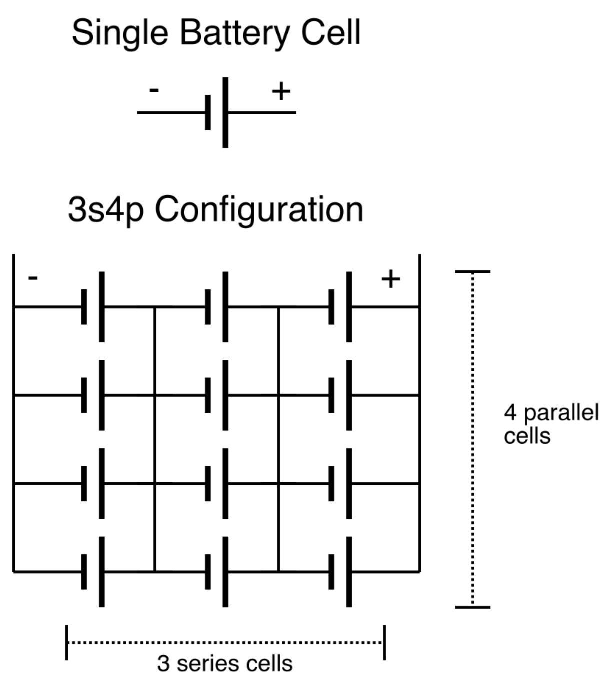
The Battery Management System (or BMS) is the hardware and firmware responsible for managing the tractive battery of Zips Racing's vehicles. This user manual details the basics of, the interacting with, and the debugging of Zips Racing's battery management systems. For technical details about a specific vehicle's BMS, refer to said project's GitHub repository.

# Tractive Battery Basics

## Battery Configurations

The tractive battery is split into multiple pieces called battery modules (or battery segments, in older text). Each of these modules is made of many individual battery cells. These cells are the smallest component of the battery, being the actual energy storage device. These cells are connected in both series and parallel with one another to increase the voltage and capacity of the module, respectively. The number of cells in series and the number of cells in parallel is often referred to as the series-parallel configuration of the battery.




For example, ZR26 has a tractive battery made of 420 cells, being 140 in series and 3 in parallel (140s3p). These cells are split across 5 modules, meaning each module is a 28s3p configuration. All example material, unless otherwise specified, will refer to ZR26's battery configuration.

## The Battery Management System

Every tractive battery, as defined by the FSAE rules, must possess a battery management system (BMS). A BMS has 2 main responsibilities:

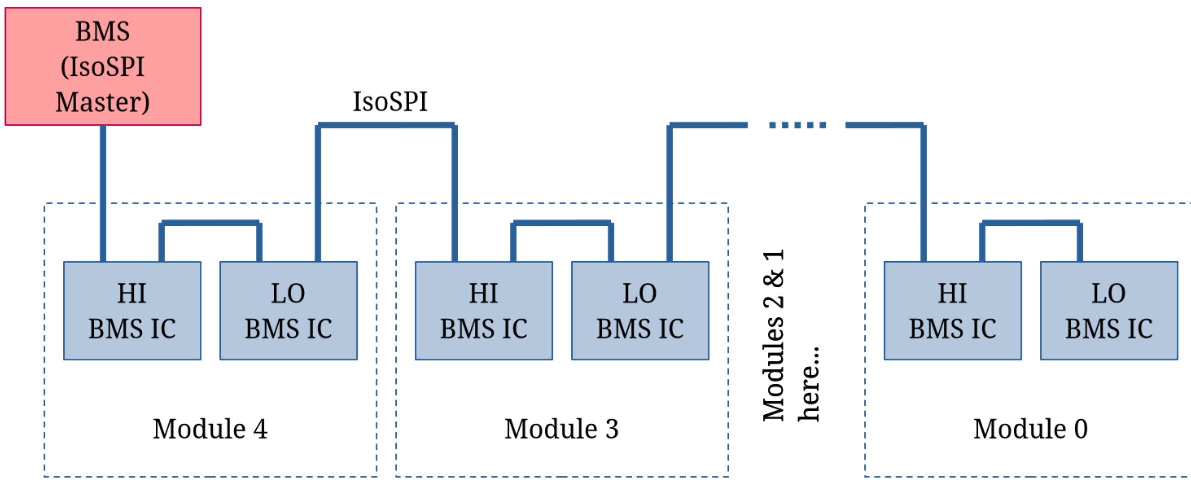
- Measuring all series cell voltages
- Measuring the temperature of 20% of all cells

Cells connected in parallel must be the same voltage, so only monitoring series cell voltages is required. When the BMS reads an invalid measurement (voltage too high or low, temperature too high), a BMS fault is asserted and the vehicle / charger must be de-energized. The thresholds at which various faults are asserted are stored in the BMS's EEPROM and thus may be changed, however care must be taken when changing such values. Overcharging or undercharging cells can lead to damage and capacity loss.

The battery management system, as it must be isolated from high-voltage components, cannot be directly connected to the cells it monitors. A family of dedicated integrated circuits (ICs) is used to solve this issue. These ICs, manufactured by Analog Devices, are often referred to as BMS ICs (or LTCs in older text).

Each module of the battery has one or more BMS ICs that are responsible for monitoring a subset of the cells in said module. For instance, the LTC6813 IC can monitor up to 18 series cell voltages and 9 temperature sensors. These ICs are connected to the BMS by a communication protocol called IsoSPI. If an error is detected in any one of the BMS ICs, or the IsoSPI communication, the BMS will assert a BMS fault.

In ZR26, each module has 2 BMS ICs. One IC, referred to as the LO IC, monitors the first 14 series cells and 5 temperatures. The other IC, referred to as the HI IC, monitors the last 14 series cells and 5 temperatures.




*For more details on the requirements of the BMS, see the FSAE rules.*

## The Isolation Monitoring Device (IMD)

A 3rd party device, called the Isolation Monitoring Device (IMD), must be installed in the tractive battery in addition to the BMS. This device is solely responsible for measuring the amount of electrical isolation between high-voltage and low-voltage components. If the degree of isolation is not sufficient (due to worn insulation, water ingress, etc.), the IMD will assert an IMD fault and de-energize the vehicle / charger.

*For more details on the IMD, see the FSAE rules.*

## Cell Sense Lines

In order to measure the voltage of a cell, the BMS IC must be connected to it by 2 wires (1 for the negative terminal, 1 for the positive terminal). The “wires” are referred to as the cell’s sense lines. In practice, these can be copper bus bars, PCB traces, or actual wires. For multiple cells in series, some of these wires can be shared (negative of one cell is the positive of the next).

Some types of BMS ICs (LTC6813 for instance) can detect if there is an issue with a sense line. If the sense line is open (broken solder joint, bad spot weld, blown fuse, etc.) the BMS IC can report an open sense-line fault to the BMS.

## Cell Balancing

Cells connected in series are not guaranteed to all be the same voltage, unlike cells connected in parallel. How close each series cell voltage is to the other series cells is referred to as the “balancing” of said cells. The difference between any series cell’s voltage and the lowest series cell voltage is referred to as the “delta” of that cell. For instance, consider 4 cells connected in series having the following voltages: $3.1V$, $3.2V$, $3.4V$, $3.1V$. The maximum voltage is $3.4V$ and the minimum voltage is $3.1V$, so the maximum delta would be $300mV$. This is a very poor balancing, in general a maximum delta of $50mV$ to $100mV$ is seen as bad.

An out-of-balance battery is not ideal, as the higher the maximum delta is, the less usable capacity the battery has. This is due to the way “full charge” and “full discharge” are defined. A battery is fully charged when the highest series cell voltage has reached the maximum of the cell type (say $4.2V$, for instance). If this cell is a higher voltage than the others due to a poor balancing, then the other cells of the battery are not fully charged. Similarly, a battery is fully discharged when the lowest series cell voltage has reached the minimum of the cell type (say $3.0V$, for instance). If this cell is a lower voltage than the others due to a poor balancing, then the other cells of the battery still have charge that could have been used. Generalized, the higher the maximum cell delta is, the less capacity the battery has.

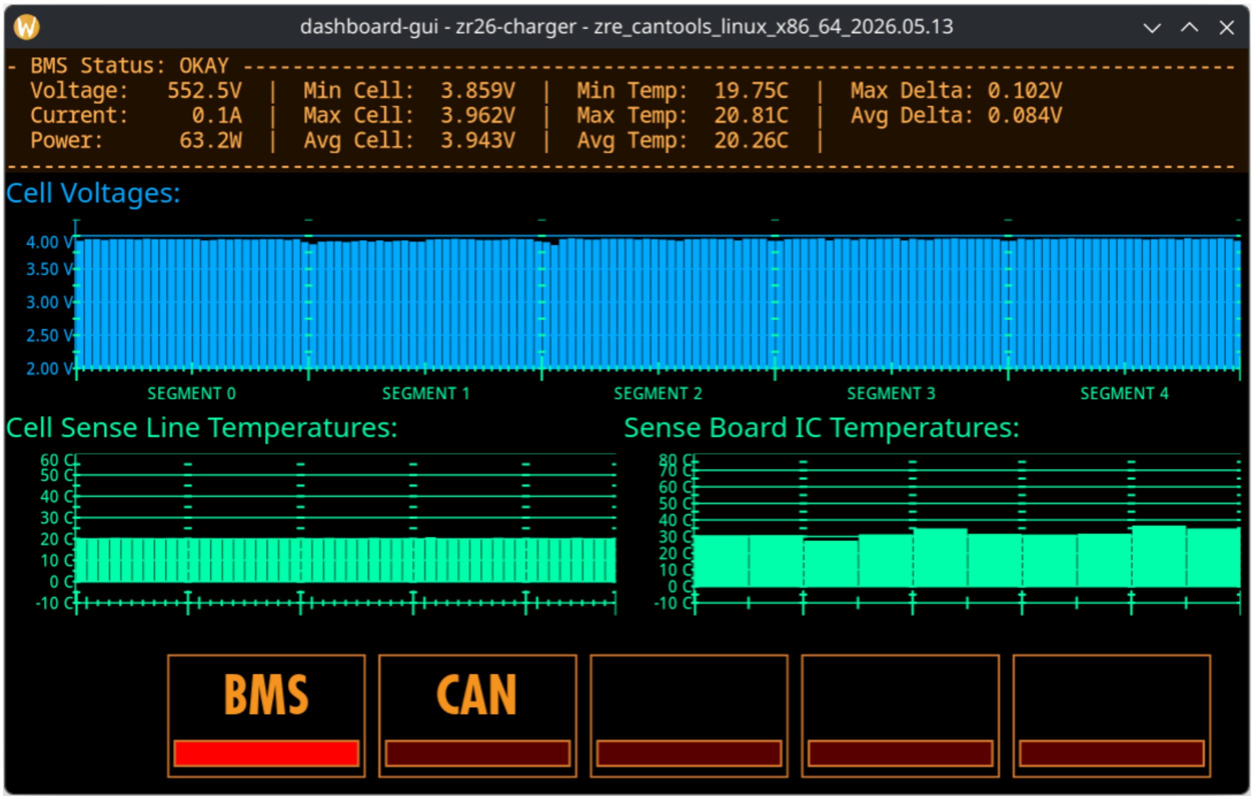
The balancing of a battery can be fixed by the BMS, via a process aptly called “cell balancing”. There are two versions of cell balancing: active balancing and passive balancing. Due to its complexity, active balancing is not employed in any of Zips Racing’s BMSs, and will therefore not be discussed here.

Passive balancing is the process of discharging cells that are too high of a voltage in order to get them closer to the minimum cell voltage. In Zips Racing's BMSs, cell balancing is automatically enabled when the charger reaches the maximum cell voltage. Any cell that has a delta greater than a pre-set threshold (say $10mV$, for instance), will be discharged until it is within this threshold. Once this process is complete, the maximum cell voltage will be no higher than said threshold, meaning this threshold determines the quality of the final balancing. The downside to a higher quality balancing is it takes more time to reach. In general, passive balancing can take 10+ hours to reach the desired threshold.

*It is important to allow the BMS time to balance after charging, as to avoid letting the battery's balancing from getting too far out-of-line. It is especially important to allow the battery to fully balance before capacity-sensitive events (such as endurance).*

## The Dashboard BMS Page

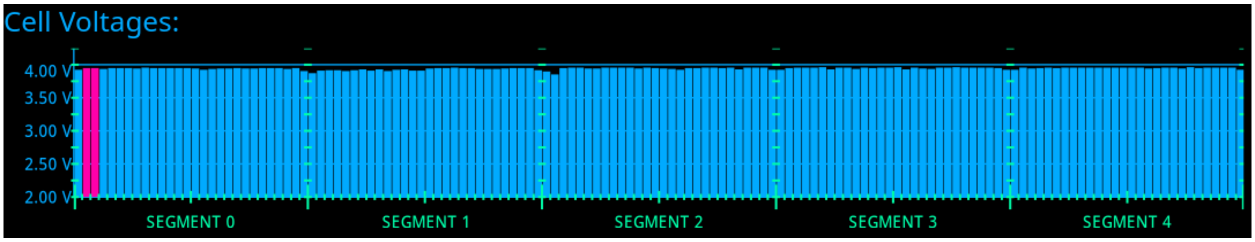
The `dashboard-gui` of ZRE-CAN-Tools, or the DART itself, may be used to monitor the status of the BMS. An example of the page for ZR26 is provided below.



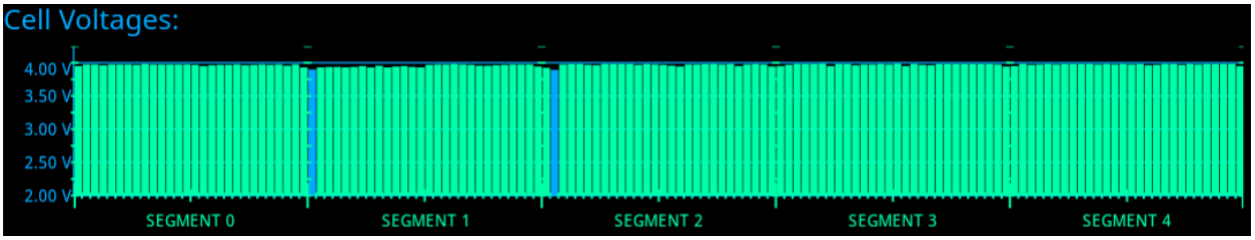

The *BMS Status* panel provides a high-level overview of the BMS. If a BMS fault is asserted, such will be indicated in the top line of the status panel. In the 1st column, the instantaneous voltage, current, and power delivery for the battery are displayed. The power and current here use the active sign convention, meaning a positive value corresponds to power being delivered by the battery to the vehicle. The 2nd and 3rd column display the minimum, maximum, and average cell voltages and sense line temperatures. The 4th column provides the maximum and average cell voltage delta (see *Cell Balancing* for more info).

The *Cell Voltages* bar graph provides the voltage of every series cell in the battery. These values, listed left-to-right, are ordered by increasing voltage potential. This means the left-most bar is the lowest potential cell and the right-most bar is the highest potential cell. The tick markers along the bottom of the graph divide the cells by battery module and BMS IC. A large tick marker denotes the division of two battery modules. A medium tick marker denotes the division of two BMS ICs.

The color of each bar indicates information about the cell. Blue cells (as seen in the example) are cells that are okay. Red cells (seen in 1st example below) indicate an invalid voltage measurement. This can be due to a variety of reasons, usually either an open sense line or expired CAN bus data. Green cells (seen in the 2nd example below) indicate a cell is being discharged in order to balance the cell voltages.




*A pair of invalid cell voltage measurements. In this case, the culprit is an open sense line.*




The cell voltage bar graph during balancing. The 2 blue cells are the only ones not being discharged, as they are the lowest 2 voltages.

The *Cell Sense Line Temperatures* bar graph provides every known sense line temperature. Like the *Cell Voltages* bar graph, these are listed in order of increasing potential, however, this graph does not provide a way to identify these. The tick markers on this graph have the same meaning as the tick markers in the *Cell Voltages* bar graph.

The *Sense Board IC Temperatures* bar graph provides every known BMS IC temperature. Like the *Cell Voltages* bar graph, these are listed in order of increasing potential. The tick markers on this graph have the same meaning as the tick markers in the *Cell Voltages* bar graph. Depending on the context, this information may not be available (when installed in the vehicle, for instance).

# Battery Indexing

To ensure that mechanical CAD, electrical schematics, and software are all consistent, this document defines the naming and indexing of components within the battery.

## Battery Modules and BMS ICs

The modules of a tractive battery are indexed in order of electrical potential. This means the negative-most module is considered the first and the positive-most module is considered the last. This index, for software-consistency, starts from 0. For instance, in ZR26, the negative-most module is referred to as “Module 0” and the positive-most module is referred to as “Module 4”.

The BMS ICs follow the same convention as the modules. Within a module, the negative-most BMS IC is the first and the positive-most is the last. An important thing to note here is that this is not necessarily the same as the IsoSPI daisy-chain ordering. It is the responsibility of the BMS firmware to convert between the daisy-chain ordering and this convention of indexing.

In BMSs that have 1 BMS IC per module, the BMS IC index and the module index are the same. In BMSs that have 2 BMS ICs per module, they may be referred to as the “LO” IC and the “HI” IC, where LO is equivalent with 0 and HI is equivalent with 1.

## Battery Cells and Sense Lines

Battery cells follow the same convention as the battery modules do. The negative-most series cell is considered the first and the positive-most series cell is considered the last. There currently is no convention for referring to individual cells, rather the collective of N parallel cells would be referred to by the series cell index.

Battery sense lines follow the same convention as the battery cells, but with a slight modification. As there can be multiple sense lines with the same electrical potential (1 in 1 module, 1 in another, for instance), the indexing must “double up” on occasion. In this case, the sense lines both have the same index and are distinguished by a “HI” or “LO” postfix. For instance, in ZR26 there are 28 series cells per module. The positive sense line of cell 27 and the negative sense line of cell 28 are both the same electrical potential, but they are two separate wires (due to being in different modules). In this example, the sense line connected to the positive of cell 27 would be referred to as “sense line 28 LO” and the negative sense line of cell 28 would be referred to as “sense line 28 HI”. See the below example if you need more clarification.

## Comprehensive Example

The example below is an (abridged) diagram of all the electrical components forming ZR26's tractive battery. Note that the ellipsis (…) is used to indicate a section has been omitted for simplicity (group of cells or battery modules). The pattern for indexing does not change inside of the omitted regions.